

Theory of Computation

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

Exercise 7: (More exercises on Pumping Lemma)

Prove that the following language is not Regular

$$L = \{ a^i b^j c^k : k < 2i + j \}$$

Short prove: Assume L is Reg Lang. $|3m-1 \quad m \quad m|$ m is the no. of states

$$W = a^m b^m c^{3m-1} ; w \in L, |w| \geq m$$

$$\underbrace{a \dots a}_{x} \underbrace{b \dots b c \dots c}_z ; |xy| \leq m, |y| \geq 1$$

let $y = a^k \quad k \geq 1$

$$i = 0, xy^0z = a^{m-k} b^m c^{3m-1}$$

$$\text{Since } k \geq 1, 3m-1 < 2(m-k) + m$$

$$3m-1 < 3m-2k$$

$\therefore$ Therefore, xy^0z is not in L

PBC.